

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

Coordinated cortical activity need not describe a single experienced situation. We develop a conditional field account that distinguishes collective timing, compatibility of local descriptions and an internal representation of the resulting state. The physical hypothesis is that changing extracellular geometry increases field-mediated coupling during awakening. A noisy mean-field phase model gives a square-root stationary onset, but finite ramps delay the response. Increasing coupling, decreasing diffusion and a matched common drive can also produce identical stationary phase distributions, so recovery measurements must resolve dynamics and distinguish the interaction mechanism. The representational problem is separate: an explicit counterexample has equal phases with incompatible local states, and a plasticity control preserves coherence while losing environmental alignment. These results show what coordination alone leaves undetermined. We propose the state glued from compatible spatial measures as a formal correlate of unity, and its accurate internal reconstruction as a correlate of a minimal self. Lean 4 checks the conditional composition under eight connecting hypotheses and additional structures.

The construction establishes uniqueness for specified local data and a contracting map; identifying those objects with meaningful neural representations remains open. The account motivates tests of how cortical coordination maintains agreement about shared content and whether an internal representation of that same state is required for unified experience.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Yet even successful coordination leaves a question: what makes the coordinated activity a description of one experienced situation, with a perspective on that situation?

We propose an account with three requirements. *Coherence* concerns whether coupling sustains collective activity. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting global state supports an accurate model of itself. The latter two express commitments about consciousness whose motivation and possible dissociations we develop below.

Consider the experience of reaching for a cup. A description of the cup's location and a description of the hand's relation to it can share information

18 about where the cup is while contributing different contents to the episode.
19 Synchronous activity in the two processes would leave any disagreement
20 about that shared quantity unresolved. The formal counterexample isolates
21 this problem using spatial mass profiles at a common phase that disagree
22 on an overlap. We return to the cup example to ask what agreement would
23 mean for perceptual content, and what an internal representation would have
24 to add to place the perceiver in that situation. It is an illustration of the
25 modelling task, not a neural model implemented here.

26 Awakening gives the physical requirement a concrete setting. We hypoth-
27 esize that changing extracellular geometry slowly increases effective coupling
28 through a synchronization threshold. Under stationary branch tracking, a
29 noisy phase model fixes a square-root onset of coherence. The experiment
30 faces two distinct obstacles. Finite ramp speeds delay the response, so a
31 fitted onset exponent depends on the drive. Moreover, successive stationary
32 phase distributions can be identical under increasing coupling, decreasing
33 diffusion or a suitably matched common drive. Recovery must therefore be
34 measured in a way that identifies dynamics and distinguishes causes, rather
35 than inferred from the shape of a coherence trace alone.

36 A plasticity control tests another route to representation. Its squared-
37 drift objective rewards equalizing drift, which in the chosen frequency-cluster
38 model favors connections across the clusters that define the diagnostic tem-
39 plate. Coherence survives while alignment with that template declines. To-
40 gether with the overlap counterexample, this result establishes a limit on

41 what the coordination model explains. Compatible local content needs its
 42 own mechanism and observable.

43 The positive proposal begins at that limit. Compatible spatial measures
 44 determine one global state; an encoding within that state can reconstruct it
 45 through a specified readout. We argue that these relations capture aspects
 46 of unity and an internal perspective, while examining why reconstruction by
 47 itself is insufficient for the psychological interpretation. Lean 4 checks the
 48 composition under explicit connecting hypotheses and additional structures,
 49 with toy witnesses showing that the assumptions can hold together. The
 50 proofs establish what follows from those choices; the main unresolved work is
 51 to identify their neural counterparts and explain how coordination maintains
 52 them. Figure 1 locates these proposed connections, and Section 6 states their
 53 physical and thermodynamic assumptions.

54 **2 Coherence: the coupling model**

55 **2.1 Dynamics and the stationary threshold**

56 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

57 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij}
 58 has inverse-time units, with angles dimensionless. The uniform mean-field

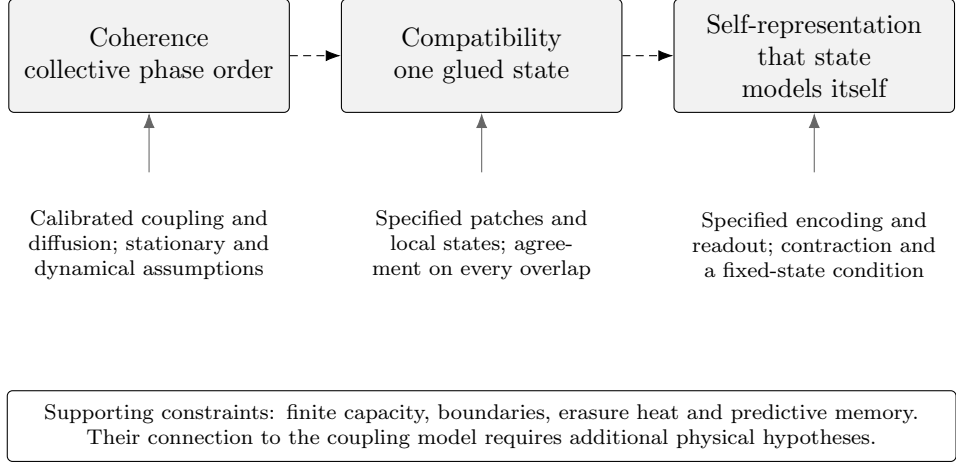


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by the formal composition.

59 convention is $K_{ij} = K/N$. Frequency heterogeneity and phase diffusion are
60 separate parameters.

61 For identical frequencies in their rotating frame, the stationary ansatz is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

62 where I_n are modified Bessel functions and ψ is the collective phase. Lean
63 verifies normalization, the circular order parameter and self-consistency for
64 this density. At $K \leq K_c = 2D$, the only nonnegative self-consistent order
65 parameter is zero. At $K > K_c$, exactly one positive solution coexists with

66 zero; it emerges continuously and increases strictly with K . Near threshold,

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

67 These results characterize the stationary branch (Fig. 2). Connecting it
 68 to finite trajectories requires a dynamical mean-field limit and dynamical
 69 selection, neither proved here. Propagation of chaos concerns the first of
 70 these steps (Sznitman, 1991); exchange symmetry alone does not establish
 71 independence, especially when realizations have an unpinned collective phase.

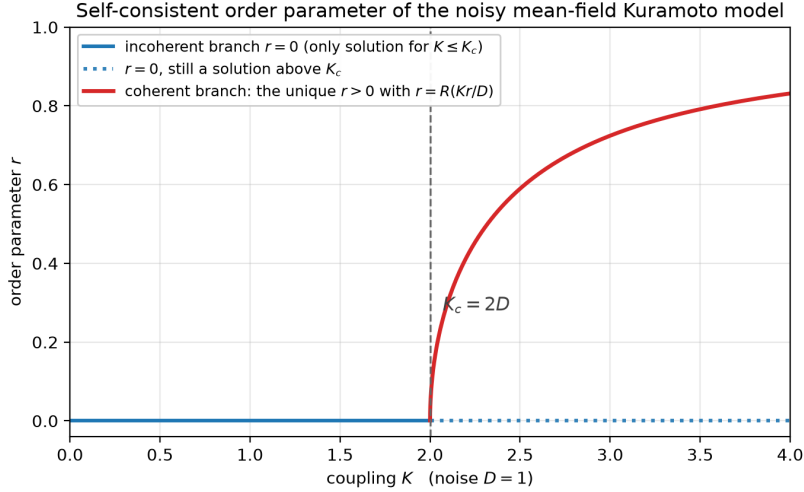


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

72 A spatial version replaces the finite sum by an integral against a kernel
 73 $K(x, y)$. The substrate measure has total mass one, making the mean-field

74 strength a kernel average. Applying the scalar threshold to a heterogeneous
 75 or signed kernel requires a justified reduction to that mean-field description.

76 **2.2 Physical identification and calibration**

77 The cortical hypothesis identifies part of the effective coupling with endoge-
 78 nous extracellular electromagnetic interactions. Fields can modulate spike
 79 timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field
 80 models have been proposed for coordinating distributed neural information
 81 (Pinotsis & Miller, 2022; McFadden, 2020). The quantitative question is
 82 whether that interaction supplies sufficient coupling relative to diffusion.
 83 Changes in phase organization under propofol (Bastos et al., 2021) moti-
 84 vate studying this regime, but do not measure the coupling itself.

85 The physical field predicate requires a jointly continuous kernel, a nor-
 86 malized substrate, a mean-field strength equal to the spatial average, and no
 87 single site providing more than half that strength. Regular mesh refinement
 88 establishes convergence of a scalar coupling-energy quadrature; constructing
 89 the kernel and relating it to trajectories are separate modelling tasks. The
 90 formal witnesses supply toy instances of the predicate.

91 Calibration is essential because a spike-time shift is not a coupling rate.
 92 An estimate proportional to $N\Delta tf$, using population size, timing shift and
 93 oscillation frequency, is dimensionless. A conversion of the form

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

94 requires a rate γ , where s is the timing shift per unit field and E the field
 95 amplitude. This rate specifies how induced timing shifts persist and accumu-
 96 late into phase drift. Supplemental Material explains why the other factors
 97 cannot determine it. The biological threshold test requires independent es-
 98 timates of the interaction kernel, its aggregation, and phase diffusion in the
 99 same cortical regime.

100 **3 Testing coherence through recovery after** 101 **awakening**

102 The proposed control variable is extracellular geometry. Sleep-associated
 103 changes in extracellular space and clearance (Xie et al., 2013) motivate a
 104 coupling parameter that evolves more slowly than phase dynamics. The
 105 extracellular volume fraction is approximately $\alpha = 0.2$ at diffusion tortuosity
 106 $\lambda \approx 1.6$ (Syková & Nicholson, 2008); the cited sleep study reports a rise
 107 in α from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz cortical
 108 tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017).
 109 In an effective-medium description at fixed transmembrane current density,
 110 a larger extracellular volume fraction raises conductivity and reduces field
 111 amplitude.

112 This gives the coupling-gated hypothesis a directional prediction: contrac-
 113 tion of extracellular space on awakening raises E and, with the other factors
 114 in Eq. (4) held fixed, raises effective coupling. The proposed geometry–

115 coupling relation and its role in human recovery both require measurement.
 116 The effective-medium argument supplies a sign under stated conditions; cal-
 117 ibration supplies the magnitude and the mapping to the scalar model.

118 If $K(t)$ rises through K_c and the phase distribution tracks the coherent
 119 stationary branch, Eq. (3) gives

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

120 The crossing time satisfies $K(t_0) = K_c$. It is a prediction of latency only when
 121 coupling is measured independently of coherence. The subsequent shape
 122 and saturation depend on the later coupling trajectory. Critical slowing
 123 down restricts branch tracking near threshold, while finite populations have
 124 fluctuating order below it. A measurable onset therefore needs an explicit
 125 criterion above that fluctuation floor.

126 **3.1 What a recovery distribution can identify**

127 Equation (2) gives a concentration–coherence relation with no fitted shape
 128 parameter. Testing it requires separate estimators: r from circular resultant
 129 length and a from density shape. Estimating a by inverting R from the
 130 measured r makes agreement automatic. Even a valid agreement tests a
 131 statistical description shared by the von Mises family; it does not determine
 132 the coupling medium.

133 There is a stronger limitation. Successive equilibrated phase distributions

134 cannot, by themselves, distinguish three different recovery mechanisms. Let
 135 $q(t)$ be a prescribed ratio crossing 2, let $r(q)$ denote its stationary branch,
 136 and choose a fixed rate $D_0 > 0$. Increasing coupling with $K(t) = D_0 q(t)$
 137 and $D = D_0$ gives the same stationary density as fixed coupling K_0 with de-
 138 creasing diffusion $D(t) = K_0/q(t)$. Both have concentration $a = q(t)r(q(t))$.
 139 Uncoupled oscillators under a common prescribed drift $-h(t) \sin(\theta - \psi)$, with
 140 $h(t) = D_0 q(t)r(q(t))$ and $D = D_0$, have that same density. Their distribu-
 141 tions coincide window by window, even though their physical causes differ.
 142 Supplemental Material supplies the construction and its numerical feasibility
 143 check.

144 The missing observable is the physical clock. Resolved phase increments
 145 contain rate information: in the ideal continuous-path model, quadratic vari-
 146 ation accumulates at rate $2D$, while conditional increment means constrain
 147 drift. This can separate coupling and diffusion under the specified observa-
 148 tion model. Distinguishing endogenous coupling from the matched common
 149 drive additionally requires a measured drive or a controlled perturbation, be-
 150 cause their mean-field drifts can coincide. Identifying the coupling as electro-
 151 magnetic then requires evidence specific to that interaction. This hierarchy
 152 determines the experiment: distribution, dynamics and physical mechanism
 153 are distinct targets.

154 **3.2 Finite ramps change the observable onset**

155 Finite simulations characterize how the stationary prediction can appear in
156 a recovery measurement. The ramp study uses $N = 2000$, $D = 1$ and 32
157 replicas at four coupling speeds. Escape is the first of three consecutive
158 saved samples at $r \geq 0.2$ after threshold. The supplement gives parameters,
159 estimators and further controls.

160 Faster ramps delay escape (Fig. 3). Only 5/32 replicas escape within the
161 fastest window. The three slower ramps have mean coupling delays 0.1947,
162 0.0684 and 0.0253. Their fitted onset exponents run from 0.898 to 0.417,
163 compared with 0.443 to 0.448 from the same estimator on the exact stationary
164 branch at the same coupling grids. At the fastest speed the ensemble mean
165 never reaches the declared fit window; the returned value 0.964 is a local
166 log-slope of a delayed trajectory.

167 This is delayed passage through a noisy bifurcation (Berglund & Gentz,
168 2002), with a direct consequence for the awakening test: the apparent expo-
169 nent depends on the ramp and on the fitting window. A deviation from the
170 square-root law constrains the adiabatic approximation only where branch
171 tracking is independently justified. The sweep does not bracket the specified
172 deviation-based critical ramp speed.

173 Additional controls test the approximations used by the physical model.
174 Time-step refinement attributes most of the sampled discrepancy from the
175 stationary order to integration bias. Population-size diagnostics show why
176 a shared random collective orientation can preserve unconditional pair cor-

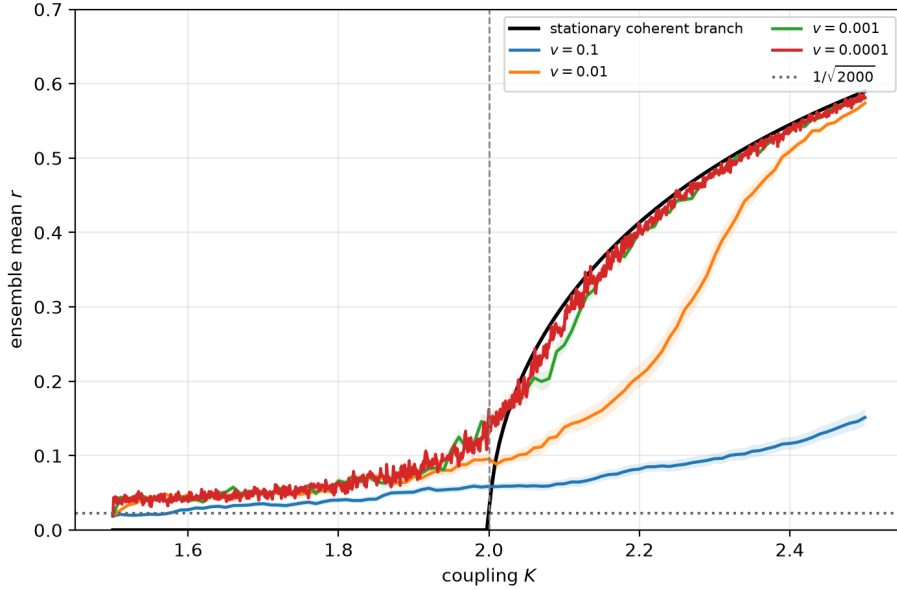


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

177 relation in the coherent regime. Spatial and signed-network controls exhibit
 178 coherence at specified total coupling, with the refined signed-network cross-
 179 ing bracket $K_{\text{eff}}/D \in [1.958, 2.009]$ still containing 2. These controls delimit
 180 finite regimes and numerical resolution; the supplement reports their full
 181 evidence. Calibrating the cortical coupling remains the experimental task.

182 3.3 What the estimators produce on EEG

183 Public propofol-sedation EEG (Bajwa et al., 2025) provides an exploratory
 184 exercise of the concentration and coherence estimators. The implemented

185 analysis pools channels over 100-ms windows, estimating a spatiotemporal
186 phase distribution. Bipolar and common-reference montages produce differ-
187 ent residuals. The bipolar results are compatible with the Bessel relation,
188 but the concentration range is too narrow to distinguish it from its linear
189 approximation at the observed scatter.

190 An informative recording needs a broader concentration range, estima-
191 tor calibration that accounts for dependence, and an observation model for
192 source mixing, referencing and temporal pooling. The sleep-recovery hypoth-
193 esis remains untested by these sedation recordings.

194 **3.4 A protocol for the awakening experiment**

195 The analysis defines six requirements for the physical test.

- 196 **1. Coupling and diffusion in the same units.** Estimate K and D
197 from resolved phase dynamics in the same cortical regime, separating
198 diffusion, frequency heterogeneity and observation noise. A measured
199 drive or controlled perturbation is additionally needed to distinguish
200 endogenous coupling from the matched common-drive alternative.
- 201 **2. Geometry calibrated to coupling.** Establish the relation between
202 an extracellular-geometry observable and effective coupling, with un-
203 certainty. A diffusion-MRI or astrocytic proxy for volume fraction re-
204 quires its own validation; the conductivity argument supplies only the
205 expected sign.

206 **3. Adequate phase measurements.** For the proposed spatial estima-
207 tor, use at least 100 sites per estimate, r from resultant length, and
208 a from unweighted regression of log density on the cosine of centered
209 phase. This numerical minimum needs null calibration at the actual
210 sample count, concentration range and dependence. Temporal pooling
211 changes the estimand.

212 **4. An independent recovery endpoint.** Measure behavioral recovery
213 separately from the phase trace.

214 **5. A predeclared onset criterion.** Fix a coherence threshold above the
215 finite-size floor and account for finite ramp speed before interpreting
216 an onset exponent.

217 **6. Matched model comparison.** Compare coupling-gated recovery
218 with diffusion change, common drive and alternative slow-recovery
219 models under the same observation and filtering assumptions. Station-
220 ary distributions alone cannot resolve the equivalence in Section 3.1.

221 Failure of the concentration–coherence relation over an informative range
222 would constrain the phase-density model. Failure of the calibrated geometry–
223 coupling association would constrain the proposed physical gate. A com-
224 peting dynamical explanation that better predicts independently measured
225 recovery would count against its role in awakening. These tests address the
226 first requirement of the framework. Even a favorable outcome leaves the
227 question of what the coordinated activity represents.

228 4 Compatibility: from coordinated activity 229 to a global description

230 4.1 Why common timing is insufficient

231 Consider overlapping patches U and V , each carrying a local description.
232 Compatibility requires that restricting either description to $U \cap V$ gives the
233 same object. Agreement concerns the shared domain; descriptions on distinct
234 parts can remain different. For the finite spatial measures used here, equal
235 overlap restrictions give a unique measure on the covered domain. This is
236 the gluing property formalized by a sheaf.

237 In the cup example, suppose a visual description locates the cup on a
238 table and a reaching description relates the hand to the cup. To compare
239 them, a model must express their shared quantity, such as cup location, in
240 a common coordinate system. Restriction would retain that quantity while
241 dropping details peculiar to each description. Agreement on it permits the
242 visual description to contribute surface appearance and the reaching descrip-
243 tion to contribute hand configuration without duplicating the whole situa-
244 tion. Disagreement after coordinate conversion would make the two estimates
245 incompatible at that level of description, however closely timed their activity.

246 The formal counterexample makes the additional requirement visible. Ev-
247 ery patch has the same oscillator phase, yet two local mass profiles assign
248 different densities to a shared site. They cannot be restrictions of a single

249 global measure. Phase order therefore cannot establish compatibility. Con-
250 tinuity of a physical field does not repair disagreement in the descriptions
251 associated with it.

252 For a theory of experience, the attraction of gluing is this contribution
253 of different local contents to one description. The cup example does not yet
254 instantiate the theorem: receptive territories and content domains must be
255 related to the spatial patches, and their descriptions must have the required
256 extension property. In particular, a decoder assigning probability laws to per-
257 ceptual contents needs a further extension model, because matching overlap
258 marginals need not determine a unique global joint distribution (Abramsky &
259 Brandenburger, 2011). The spatial-measure construction provides a precise
260 case of gluing; a content model must justify its own existence and uniqueness
261 claims.

262 **4.2 Coherent adaptation can lose environmental align-** 263 **ment**

264 An adaptive coupling rule might appear to supply that missing representa-
265 tional structure. The squared-drift control shows why the objective mat-
266 ters. Symmetric coupling conserves mean drift: pairwise sine terms cancel
267 when summed over nodes. Minimizing the sum of squared drifts at that
268 fixed mean therefore rewards equalizing drift across nodes. In the chosen
269 frequency-cluster model, this favors coupling between unlike-frequency clus-

ters, whereas the diagnostic template rewards coupling within like-frequency clusters. The objective and the scored structure pull in different directions.

For fixed phases, the continuum objective is

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

The formalization constructs the bounded drift operator A and its adjoint, obtains $\nabla\sigma = (2/D)A^*(AK + \omega)$, and proves descent for the specified gradient flow. This is a squared-drift functional; identifying thermodynamic entropy production would require probability-current information.

The finite control co-evolves noisy phases and symmetric, nonnegative couplings at fixed total resource. The frequency-cluster template is a diagnostic covariance of shared latent signals, not a training input. Across three seeds, second-half order remains between 0.9167 and 0.9222, while the within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4). The supplement reports random-update, relabelled-gradient and frozen-kernel controls, including comparisons at overlapping deformation and coherence ranges and against partitions blind to frequency.

The objective reduction is transient: averaging over complete update intervals removes the apparent gain seen immediately after updates. A pre-specified grid of 6 rate–cadence combinations contains no case meeting the declared 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified

290 structure, with no demonstrated sustained cost reduction.

291 Kernel alignment and overlap compatibility are different observables.
292 This control tests the narrower suggestion that coherent adaptation under
293 this objective acquires the designated environmental structure. It fails for
294 an intelligible reason: the objective rewards a different organization. A
295 thermodynamically specified learning model can include data in its objective
296 (Corberi et al., 2026); explaining agreement about the cup would likewise
297 require a learning rule connected to the relevant content.

298 **4.3 What must be specified and measured**

299 Compatibility enters the construction as a hypothesis about the local states.
300 For deterministic positive coupling, convergence to a potential minimum is
301 proved under an explicit initial basin condition; the proposed bridge E78
302 connects supercritical noisy coherence to relaxation of the specified cover.
303 Neither step explains how initially inconsistent representations become con-
304 sistent. That is the mechanism a biological account must add.

305 A neural test must first identify the domains and objects whose agreement
306 is at issue. Two decoders should estimate a shared content on overlapping
307 domains, with restriction maps specified before inspecting their agreement.
308 Their discrepancy needs a null under decoding noise, and the procedure must
309 distinguish deliberately constructed compatible and incompatible examples.
310 A common prior can make separate decoders agree artificially; region-specific
311 bias can make them disagree artificially. The supplement develops these re-

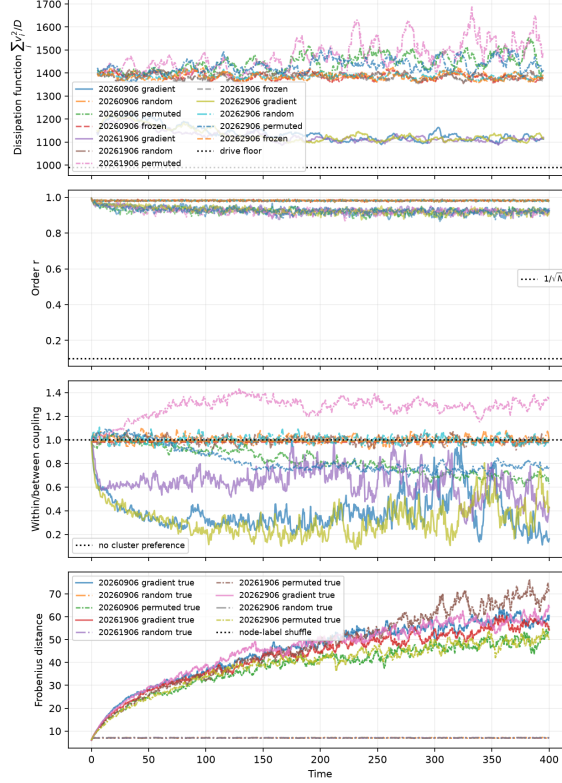


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

312 quirements. An estimator built from phase agreement would miss the prop-
 313 erty the counterexample separates.

314 Choosing a cover is also part of choosing a candidate subject. Trivial
 315 descriptions can agree while carrying little content, and a suitably chosen
 316 partition can conceal disagreement. The scientifically useful choice must
 317 therefore be constrained independently by what local activity encodes and
 318 which shared quantities can be compared. The uniqueness theorem then
 319 answers a precise question about that choice: whether the supplied local
 320 descriptions determine one global state. It leaves the empirical selection of
 321 domains and descriptions to the model.

322 **5 Self-representation: a state that models it-** 323 **self**

324 **5.1 The reflexive construction**

325 Compatibility yields a global state. Reflexivity adds a relation between that
 326 state and an encoding within it. Equip global sections with a complete
 327 metric. Under the additional restriction-resonance condition, the encoding
 328 of a section s is its restriction to a region U , and the map takes the form

$$F(s) = \text{readout}(s|_U). \quad (7)$$

329 The same region and readout are used across the model’s state space. If F
 330 contracts and fixes the section s_* supplied by the cover, Banach’s theorem
 331 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
 332 plies the contraction law and $F(s_*) = s_*$ for the boundary’s encoding–readout
 333 composite. Identifying that encoding with restriction is a separate resonance
 334 condition; E89 does not supply it. The supercritical order parameter pro-
 335 vides an inequality used by the specified contraction bound; constructing the
 336 encoding and readout remains a separate obligation.

337 The equality ties the reconstruction to the state formed from the local
 338 descriptions. In the cup example, the candidate global state would include
 339 the represented scene and the perceiver’s relation to it. A local encoding
 340 would have to retain enough information for the specified readout to recover
 341 that modelled state. An encoding of cup position alone would be insufficient
 342 if the readout could not recover the represented hand and viewpoint. The
 343 term *predictive* names this reconstruction map; a temporal neural prediction
 344 requires a dynamical interpretation.

345 **5.2 Why connect these properties to unity and a min-** 346 **imal self?**

347 We identify the section glued from compatible local states with the unity of
 348 a conscious episode, and its accurate internal reconstruction with a minimal
 349 self. The motivation concerns two features of the episode: different contents

350 belong to one situation, and that situation is experienced from a perspec-
351 tive. The mathematical relations give these proposals a definite form; their
352 psychological interpretation needs a further argument.

353 The unity proposal starts from a distinction between having several con-
354 tents and having several experiences. Seeing the cup's surface and feeling
355 the reaching hand contribute different contents to one situation. Compat-
356 ibility expresses their agreement about what they share while preserving
357 those differences. An episode can also contain uncertainty or conflicting
358 judgments. A model might represent the cup's apparent position separately
359 from a judged position and consistently describe the discrepancy. Deciding
360 whether two estimates concern the same quantity or distinct aspects of ex-
361 perience is therefore part of defining the content model. A choice made only
362 to ensure agreement would give the unity proposal no empirical force.

363 The motivation for a minimal self is that the cup is encountered from a
364 location and in relation to a body that can reach it. The proposed internal
365 representation concerns the perceiver's condition in that encounter: where
366 the perceiver is situated, what is being represented and how the body relates
367 to it. Such a representation could be implicit in the organization of activity;
368 it need not describe a person verbally or carry an autobiographical narrative.
369 Requiring it to reconstruct the same global state links the perspective to the
370 episode whose contents it organizes.

371 This motivates the reflexive condition but does not make accurate recon-
372 struction sufficient for a perspective. A recording of a scene can be recon-

373 structured, and a local signal can covary with a body’s position, without either
 374 being used by the system as a representation of its own condition. The pro-
 375 posed interpretation needs evidence that the encoding participates in how
 376 the system distinguishes changes in itself from changes in its surroundings.
 377 In the cup example, changing the encoded hand or viewpoint while holding
 378 the external scene fixed should have consequences for the system’s estimates
 379 of its relation to the cup that follow from the proposed readout. This is a
 380 criterion for assigning self-representational meaning to the variables; it is not
 381 a consequence of $F(s_*) = s_*$.

382 The fixed-point construction contributes a consistency requirement to
 383 that interpretation: the internal representation must be accurate about the
 384 very state that contains it. Its uniqueness prevents ambiguity for the chosen
 385 map, but does not establish that the representation is used, that it specifies
 386 a perspective, or that a perspective is experienced. A biological model must
 387 connect the encoding to those functions before the equality becomes evidence
 388 about a minimal self.

389 Accuracy is a demanding idealization. The construction concerns the
 390 variables represented by s_* , not an exhaustive internal copy of every mi-
 391 croscopic physical fact. A cortical realization would have to specify those
 392 variables and justify the fidelity needed for the identification. Approximate
 393 self-models, errors and changing states may require a weaker condition than
 394 an exact fixed point. Such a condition is a research question rather than a
 395 consequence of the theorem.

396 The reason to propose reflexivity as necessary is the thought that even an
397 unreflective episode has a perspective: the cup appears somewhere relative
398 to the perceiver, whether or not the perceiver attends to themselves. Moving
399 from that feature of experience to a requirement for internal self-encoding
400 is the philosophical hypothesis. Gluing alone does not require it. Unified
401 experience without the proposed encoding would count against the reflexive
402 requirement while leaving the compatibility proposal assessable on its own.

403 **5.3 Which cover and which self-model?**

404 Both uniqueness results are conditional on their objects. Gluing gives one
405 global extension of specified compatible local data; Banach gives one fixed
406 point of a specified contracting map. These solve ambiguity within a chosen
407 construction. They do not choose between different covers, different encod-
408 ings or different candidate subjects. Integrated information theory makes
409 exclusion an explicit part of its account (Tononi et al., 2016); the present
410 uniqueness results do not settle that selection problem.

411 The relevant cover and map must be tied to neural organization by evi-
412 dence independent of the desired conclusion. Local descriptions should pre-
413 dict the contents attributed to their domains, and their restrictions should
414 preserve meaning on shared domains. The encoding and readout should be
415 specified before testing them, with their sensitivity to changes in the pro-
416 posed state measured over a declared range. Exact reconstruction at the
417 candidate state is the fixed-point condition; fidelity during changes requires

418 its own assessment.

419 Contraction and a fixed point are easy to obtain in an uninformative
420 construction. A constant readout returning one predetermined state satis-
421 fies those conditions. Restriction makes the source of the encoding explicit,
422 but a readout can still discard what it receives. The cup example would
423 require sensitivity to the represented hand and viewpoint, and the interven-
424 tion described above would test whether that sensitivity has the proposed
425 role. Mathematical existence alone cannot distinguish a self-model from a
426 reconstruction chosen by an observer.

427 5.4 Dissociations that would bear on the identification

428 A test of the full commitment must measure its clauses separately: effective
429 coupling relative to diffusion, overlap compatibility, and the accuracy of the
430 proposed self-encoding, alongside an independent endpoint for experience.
431 A case meeting all three physical and representational conditions with evi-
432 dence against a unified conscious episode would challenge their sufficiency.
433 Unified experience with demonstrably incompatible local descriptions would
434 challenge the unity identification. Unified experience with compatibility but
435 without the specified self-encoding would challenge the reflexive requirement.

436 Anesthetic emergence, dreaming and dissociative states are candidate set-
437 tings because the framework asks whether coordination, unified content and
438 self-representation can separate. Interpreting such evidence requires a de-
439 clared relation between the experience endpoint and report; absence of report

440 alone cannot define absence of an episode. The compatibility observable in
 441 the supplement is a specification, and the corresponding neural self-encoding
 442 has yet to be identified. These are the main obstacles to testing the inter-
 443 pretive commitment.

444 The proposal concerns which state is conscious. It supplies no explanation
 445 of why satisfying these relations should be accompanied by experience. Nor
 446 do its gluing and fixed-point conditions select an electromagnetic substrate:
 447 the cortical field commitment resides in the physical identification and cali-
 448 bration. Their multiple realizability leaves the status of other biological and
 449 artificial implementations open.

450 **6 Thermodynamic constraints and the con-** 451 **ditional composition**

452 An implementation of the proposed representations must retain information
 453 and update it through physical processes. The supporting results bound
 454 finite-register capacity by $H(p) \leq \log |X|$ and erasure heat under a speci-
 455 fied system–bath model (Landauer, 1961; Norton, 2005; Shenker, 2000). A
 456 separate field-boundary result concerns a continuous field reaching discon-
 457 nected vacuum components (Kibble, 1976). These constrain one proposed
 458 architecture; the stationary phase threshold and the representational argu-
 459 ment do not depend on deriving that architecture from finite capacity. The
 460 supplement develops these results and their witnesses.

461 The constraint most directly relevant to representation concerns predic-
 462 tive memory. For a state X_t driven by an environmental signal S_t ,

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (8)$$

463 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, the bracket
 464 is nonnegative (Still et al., 2012). At a given dissipation budget, the bound
 465 limits retained information that fails to predict the next signal. Zero retained
 466 information can satisfy it, so it supplies no incentive to learn the cup’s loca-
 467 tion or maintain an accurate self-model. Connecting this budget to Landauer
 468 heat requires the nonpredictive information to be bounded by erased entropy
 469 at the same thermal scale and heat budget. Capacity, erased entropy and
 470 predictive information thus impose different constraints (Rosas et al., 2026);
 471 content and learning dynamics must still be specified.

472 6.1 What the composition establishes

473 Table 1 lists the eight implications accepted by the theorem **chain**. They
 474 connect the supporting constraints to the phase model and its proposed rep-
 475 resentational role. Each connection is a hypothesis; in particular, the finite
 476 register and the field boundary are independent starting objects.

477 Two structures also enter the theorem: a thermodynamic cover, including
 478 overlap compatibility, and a reflexive boundary specifying the local encoding
 479 and readout. Given these and the connecting hypotheses, **chain** concludes

Connection	Status	Required assumption
E12	Independent physical premise	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity and the boundary are independent roots.
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34	Bridge assumption	Erase supplies a predictive structure with the same thermal scale and heat budget, and nonzero wasted memory.
E45	Modelling assumption	The predictive dissipation bound places coupling energies in a convergent coarse-graining regime.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence drives a supplied cover to its relaxed equilibrium at coupling at least the named mean-field value.
E89	Modelling assumption	A contracting map mediated by local self-encoding fixes the section glued on that same cover.

Table 1: Connecting hypotheses of the formal composition. The cover’s compatibility is supplied as a structure field. Identifying the boundary’s encoding with restriction requires a separate resonance condition. The mathematical objects and node-level results are mapped in Supplemental Material, Table S1.

480 `UnifiedSelf`: the section glued on that cover is the unique fixed point of self-
481 prediction. Restriction resonance, needed for the interpretation in Eq. (7),
482 is a separate condition. The result binds the fixed point to the specified
483 local states; compatibility and accuracy at that state are premises, while
484 uniqueness follows from the construction.

485 The joint witness combines a double well, a register, a uniform-grid mesh
486 and a three-site cortex, establishing that the hypotheses can hold together
487 in distinct toy components. It supplies no common biological mechanism for
488 them. Counterexamples expose the role of individual assumptions, including
489 overlap agreement. The development declares no additional Lean axioms
490 beyond the foundational axioms used by its libraries; physical premises are
491 explicit hypotheses and model fields.

492 7 Discussion

493 The strongest results concern what a coherence measurement leaves unre-
494 solved. Different recovery mechanisms share stationary distributions; equal
495 phases permit incompatible local descriptions; and the tested plasticity ob-
496 jective sustains coherence while moving away from its diagnostic structure.
497 These findings make the coupling hypothesis more demanding to test and
498 prevent a successful synchronization model from doing unexplained repre-
499 sentational work.

500 The proposed account takes a further step by identifying a compatible,

internally represented state with a conscious episode. The cup example illustrates its appeal: descriptions of appearance and action can contribute to one situation, with the perceiver represented within it. The main explanatory gap is the mechanism that would maintain those relations. Compatibility is assumed, and the map’s accuracy at the glued state is assumed. The proofs establish their consequences and joint realizability in toy components; they do not yet explain why cortical activity should acquire or preserve them.

The most informative next step for that connection is a test of whether coordination helps maintain independently measured content agreement. After validating separate decoders and their overlap restrictions, a controlled change in coupling or coordination could test whether agreement deteriorates in the predicted way, with controls for changes in sensory information and decoder quality. Such a result would bear directly on the proposed role of coherence. In parallel, the awakening hypothesis needs its own calibrated geometry–coupling relation and resolved phase dynamics to distinguish a coupling increase from diffusion change or common drive. A favorable recovery trace alone would leave the representational connection open.

Testing the self clause requires a further choice: a candidate encoding whose contribution to representing the system’s own condition can be perturbed. The readout should predict which aspects of the represented scene, body and viewpoint change when that encoding changes. Evidence that it merely reconstructs data would leave the perspective interpretation unsettled; evidence of its proposed functional role would make dissociations from

524 unified experience informative. The exact fixed-state idealization must also
525 be assessed against the fidelity and time dependence of that biological pro-
526 cess.

527 Integrated-information, global-workspace and predictive-processing ac-
528 counts organize unity, access and modelling differently (Tononi et al., 2016;
529 Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal
530 contributes a candidate interaction mechanism and explicit relations to test
531 between local content and internal self-encoding. Its explanatory value will
532 turn on whether those relations can be identified independently and shown
533 to matter for experience. For the cup example, that means explaining how
534 the system maintains one represented situation and how the perceiver’s place
535 within it contributes to the episode.

536 **Data and code availability**

537 The Lean development, simulation scripts, tests and saved numerical sum-
538 maries are maintained in the manuscript source repository. Publication sim-
539 ulation values are regenerated from saved summaries without rerunning pro-
540 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
541 ds005620 (Bajwa et al., 2025); preprocessing and selection are described
542 in Supplemental Material. The source repository contains the reproducible
543 analysis and does not redistribute the raw recordings.

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547 **Use of AI tools**

548 AI assistance to this work was substantial and is not separable by task.
549 Several coding assistants and chat interfaces were used in an interleaved way
550 throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude
551 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code
552 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude
553 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them
554 they contributed to the Lean development, the simulation and analysis code,
555 and the drafting and revision of both documents. Because the tools were
556 used in combination and in alternation, no section, proof, figure or result is
557 attributable to any one of them, and none is claimed to be. The author is
558 responsible for all content, including everything a tool produced. Machine
559 checking establishes the stated Lean results under their hypotheses; it does
560 not validate biological interpretations or replace scientific review of generated
561 code and prose.

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